



Data Science Capstone Project

Project Title

Credit Risk Analysis System

Project Overview

The Credit Risk Analysis System is a data-driven application that helps financial institutions evaluate whether a loan applicant is likely to repay a loan or become a credit risk. The system analyzes customer data and predicts loan approval risk using data analysis and machine learning techniques.

This project gives students practical exposure to how banks and financial companies use data science for decision-making.

Project Objective

Develop a data science application that can:

- Analyze customer financial data
 - Identify low-risk and high-risk applicants
 - Predict loan approval probability
 - Generate risk insights through dashboards
 - Visualize customer risk patterns
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Difficulty Level

Intermediate Level

Estimated Difficulty: 6/10

Suitable for students with basic knowledge of:

- Python
- Data Analysis
- Machine Learning basics
- SQL and Visualization tools





Industry Relevance

This project is inspired by real-world systems used in:

- Banks
- NBFCs
- FinTech companies
- Credit card companies
- Loan approval platforms

Students will understand how data science supports financial risk management.

Project Modules

Module 1: Data Collection & Preprocessing

- Import customer loan dataset
- Handle missing values
- Remove duplicates
- Encode categorical data
- Normalize important fields

Learning Outcomes

- Data cleaning
 - Data preprocessing
 - Handling real-world datasets
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Module 2: Exploratory Data Analysis

Analyze:

- Customer income patterns
- Loan amounts
- Credit history
- Default trends
- Risk distribution

Create graphs and charts for insights.





Learning Outcomes

- Data visualization
- Pattern identification
- Business understanding

Module 3: Credit Risk Prediction Model

Build a machine learning model to classify:

- Low-risk applicants
- High-risk applicants

Suggested algorithms:

- Logistic Regression
- Decision Tree
- Random Forest

Learning Outcomes

- Classification models
- Model training and testing
- Prediction systems

Module 4: Risk Score Dashboard

Display:

- Risk category
- Approval probability
- Customer analysis
- Loan trends

Learning Outcomes

- Dashboard development
- Business reporting
- Data storytelling





Module 5: Final Recommendation System

Generate simple recommendations such as:

- Approve Loan
- Review Manually
- Reject Application

Based on model prediction and risk score.

Learning Outcomes

- Decision-making systems
- Practical ML applications

Recommended Tech Stack

Programming

- Python

Libraries

- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- Seaborn

Dashboard Tools

- Streamlit or Power BI

Database

- MySQL (Optional)

Deployment

- Streamlit Cloud or Render





Project Workflow

1. Dataset Collection
2. Data Cleaning
3. Exploratory Data Analysis
4. Model Training
5. Risk Prediction
6. Dashboard Visualization
7. Final Recommendation Generation

Dataset Suggestions

Students can use:

- Kaggle Credit Risk Dataset
- Loan Prediction Dataset
- Banking Customer Dataset

Deliverables

Students must submit:

- Source Code
- Project Report
- PPT Presentation
- GitHub Repository
- Dashboard Screenshots
- Deployment Link
- Demo Video

Scope Limitations

Students should avoid:

- Deep Learning models
- Real banking integrations
- Complex financial calculations
- Large-scale cloud systems

Focus should remain on building a clean and functional prediction system.





25-Day Project Plan

Days	Tasks
Day 1–3	Requirement Analysis and Dataset Collection
Day 4–7	Data Cleaning and Preprocessing
Day 8–12	Exploratory Data Analysis
Day 13–17	Machine Learning Model Development
Day 18–21	Dashboard Development
Day 22–23	Recommendation System
Day 24	Testing and Bug Fixing
Day 25	Deployment and Final Presentation

Evaluation Criteria

Criteria	Weightage
Functionality	30%
Data Analysis	20%
Machine Learning Model	20%
Dashboard/UI	15%
Deployment	10%
Documentation	5%

Final Outcome

By completing this project, students will learn:

- Real-world data preprocessing
- Data visualization techniques
- Machine learning classification
- Financial risk analysis basics
- Dashboard development
- End-to-end data science workflow

The final system will function as a Credit Risk Analysis platform capable of analyzing customer data, predicting credit risk, and generating loan approval recommendations.

